

MLOps Assignment 2 Final Report

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Course: MLOps | PGD AI Program | IIT Jodhpur
Project: Hugging Face Fine-Tuning, Experiment Tracking & Model Deployment

1. Model Selection Rationale

In this project, DistilBERT was selected as the pre-trained transformer model for text classification. DistilBERT is a lighter and faster version of BERT that maintains strong performance while reducing computational requirements. It is well-suited for Kaggle GPU environments and provides efficient fine-tuning for NLP tasks such as genre classification. The model was loaded using Hugging Face Transformers and fine-tuned on the Goodreads reviews dataset.

2. Kaggle Training Setup

The training workflow was implemented using a Kaggle Notebook with GPU acceleration enabled. Kaggle Secrets were used to securely store WANDB_API_KEY and HF_TOKEN credentials. Internet access was enabled to download Hugging Face models and upload trained artifacts. Weights & Biases (W&B;) was integrated for experiment tracking and monitoring.

3. Training Workflow

The dataset was tokenized using DistilBertTokenizerFast with truncation and padding applied. A custom PyTorch dataset class was used for training and evaluation. The Hugging Face Trainer API was configured with TrainingArguments including logging, evaluation, and checkpoint saving. W&B; automatically tracked training loss, validation loss, accuracy, and F1-score throughout training.

4. Evaluation Results

After training, the model was evaluated on the test dataset. Metrics including evaluation loss, accuracy, and weighted F1-score were recorded. A classification report was generated and uploaded to W&B; as an artifact for reproducibility and experiment tracking.

5. Results Table

Replace the placeholder values below with your actual evaluation metrics from W&B; or trainer.evaluate().

Metric	Score
Accuracy	0.58437
F1 Score	0.58095
Eval Loss	1.16838

6. Challenges & Learnings

During the implementation process, package compatibility issues and GPU memory limitations were encountered. These were resolved by installing compatible versions of transformers and huggingface_hub and reducing the training batch size when necessary. The project provided practical understanding of the complete MLOps workflow including experiment tracking, reproducibility, model deployment, and cloud-based GPU training.

7. Project Links

GitHub Repository: <https://github.com/g25ait2143-spec/mlops-assignment-2>

Kaggle Notebook: <https://www.kaggle.com/code/palaganisaichaitanya/notebook97772a1f61>

Weights & Biases Dashboard: [https://wandb.ai/g25ait2143-iitj/mlops-assignment2/runs/a6gt5pfq?](https://wandb.ai/g25ait2143-iitj/mlops-assignment2/runs/a6gt5pfq?nw=nwuserg25ait2143)

[nw=nwuserg25ait2143](https://wandb.ai/g25ait2143-iitj/mlops-assignment2/runs/a6gt5pfq?nw=nwuserg25ait2143)

Hugging Face Model: <https://huggingface.co/chaitanyapalagani/distilbert-goodreads-genres>